

Region Segmentation based on Fluorescent Imaging

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Abstract

001 *Hospital-associated saliva deposition can seed high-risk*
002 *pathogens across shared surfaces. This work studies un-*
003 *supervised segmentation of saliva deposits illuminated un-*
004 *der UV-C, where only ten color images are available and*
005 *ground truth masks only cover four centralized patches per*
006 *frame. We explore a hierarchy of methods: a MATLAB-*
007 *based luminance/thresholding pipeline (Method A), an ex-*
008 *ploratory k-means segmentation (Method C), and our prob-*
009 *abilistic Gaussian mixture model (Method D) that reuses*
010 *k-means centers, runs ten EM initializations per image, and*
011 *selects the most likely fit by pseudo-log-likelihood. The re-*
012 *sulting GMM probability maps are binarized with morpho-*
013 *logical cleanup and evaluated on accuracy, precision, re-*
014 *call, and Dice; the $K=4$ model achieves the highest Dice*
015 *while maintaining recall above 0.96. Our comparison high-*
016 *lights how converting hard clusters into likelihood-ranked*
017 *mixtures improves recall while still recovering reasonable*
018 *precision, providing a practical unsupervised pipeline for*
019 *low-data airborne contamination imagery.*

020 1. Introduction

021 1.1. Problem Definition

022 Hospital-acquired infections (HAIs) pose a recurring threat
023 to healthcare environments, increasing patient risk, compli-
024 cating workflows, and incurring higher operational costs.
025 HAIs spread by direct contact, droplets, aerosols, and con-
026 taminated surfaces. The problem is further harmful in
027 busy clinical areas, where equipment is moved frequently,
028 and patients are vulnerable. Major reviews and guidelines
029 identify environmental reservoirs, device-related contami-
030 nation, and workflow pressures as the primary factors con-
031 tributing to disease propagation in hospitals, suggesting that
032 broad infection-control policies alone are insufficient. Ad-
033 ditionally, knowing where contamination actually accumu-
034 lates is crucial, allowing for focused and effective cleaning
035 [1, 9, 11, 15, 22].

036 Saliva is a common target for efforts aimed at reduc-

ing HAIs because it carries many pathogens and is easily
spread during talking, coughing, and routine procedures.
Once saliva lands on instruments or surfaces, its appear-
ance changes as it dries or interacts with different materials,
making it difficult to spot under normal lighting. Studies
using fluorescence reveal clear differences between moist
and dried saliva on various surfaces, indicating that opti-
cal imaging can detect deposits that would otherwise be
missed [13, 23]. Other reviews and studies that treat saliva
as a diagnostic and surveillance fluid also highlight its role
in pathogen carriage and transmission, reinforcing the idea
that detecting saliva specifically is a useful way to reduce
HAIs [3, 12, 21].

In practice, hospitals employ a layered approach to pre-
vention, utilizing masks, face shields, and routine cleaning
and sterilization. Masks and shields reduce the spread of
droplets but do not eliminate surface deposition. Experi-
ments that use fluorescent surrogates often reveal contam-
ination patterns invisible to the naked eye, demonstrating
that barriers and cleaning must work together [20]. Mask
effectiveness depends on fit, material, and how long it is
worn. Leaks and saturation can still let droplets escape
and contaminate nearby surfaces [2]. Face shields block
splashes and large droplets but do not prevent peripheral
deposition, and the shield surfaces themselves can become
contaminated. Therefore, they are most effective when used
in conjunction with masks [2, 20]. Routine disinfection
lowers surface bioburden, but actual coverage and contact
time vary in real settings. Laser-induced fluorescence (LIF)
can map where droplets land and persist, providing useful
feedback to improve cleaning patterns and identify missed
spots [23].

Computer vision can make these imaging findings ac-
tionable. Segmentation algorithms can produce binary
masks that mark saliva or fluorescent surrogates in images.
These masks can guide automated or semi-automated clean-
ing tools to focus on contaminated regions. Fluorescence
imaging and high-speed visualization provide the contrast
and ground truth necessary to train and validate these al-
gorithms across various surface types and moisture states,
helping bridge the gap between lab measurements and real-

078 world cleaning workflows [2, 13, 20, 23].

079 1.2. Literature Review

080 Medical image segmentation is the area of medical tech-
081 nology that most resembles the creation of binary masks
082 for saliva recognition. This process labels every pixel (or
083 voxel) in an image to isolate organs, lesions, fluids, usu-
084 ally by difference in contrast. For contamination detection
085 this usually boils down to a binary mask (contaminant vs.
086 background), but the goals are the same as other clinical
087 tasks: get accurate boundaries, be robust to different imag-
088 ing conditions, and produce outputs that can drive decisions
089 or automated actions. Standard overlap metrics such as the
090 Dice Coefficient and Intersection over Union (IoU) are used
091 to judge how well a model matches expert masks, called
092 Ground Truths (GT). These metrics guide model selection
093 and tuning [18].

094 The methods for segmentation range from image-
095 processing techniques to deep neural networks. Basic steps,
096 such as contrast enhancement, thresholding, edge filters
097 (Sobel/Laplace), and morphological cleanup, are still used
098 for boosting visibility of small or low-contrast deposits, as
099 well as for preprocessing images before applying the model
100 [5, 18]. On the learning side, U-Net and its many variants
101 are the baseline because they combine multi-scale context
102 with precise localization. More recent work adds attention
103 blocks or mixes CNNs with transformers to capture longer-
104 range context and improve small-object detection [7, 17].
105 There’s also a growing push to use synthetic data and gen-
106 erative models to expand training sets, and public codebases
107 for droplet tasks (e.g., Droplet-ML) make it easier to repro-
108 duce experiments and try augmentation strategies [16, 19].

109 The main challenges for saliva/droplet segmentation are
110 familiar in the computer vision field. Getting enough la-
111 beled masks is expensive, droplets are tiny so class imbal-
112 ance is severe, and appearance changes a lot with surface
113 type, lighting, and drying state. That means the necessity
114 of specialized datasets, multi-scale supervision, and care-
115 ful preprocessing or domain-adaptation to generalize from
116 lab images to real clinical scenes [7, 17]. Practically speak-
117 ing, the best routes adopt a hybrid approach, using classical
118 enhancement to make deposits visible [18], training robust
119 segmentation networks (with attention to small-object han-
120 dling) [7], and relying on synthetic or open datasets to fill
121 gaps [16, 19].

122 1.3. Contributions

123 In this work we deliver:

- 124 • A structured comparison of the existing MAT-
125 LAB/thresholding pipeline (Method A), the exploratory
126 k-means clustering pipeline (Method C), and the prob-
127 abilistic Gaussian mixture model pipeline (Method
128 D).

- A GMM workflow that seeds EM with k-means centers, runs ten initializations per image, selects the best solution via pseudo-log-likelihood, and thresholds the resulting probability map before morphological cleanup.
- Empirical evidence that the $K = 4$ GMM increases Dice while keeping recall above 0.96, illustrating how likelihood-aware model selection improves the unsupervised segmentation of saliva under UV-C.

2. Dataset and Methods

2.1. Dataset description

The dataset contains ten UV-C images of saliva deposits on five substrates (aluminum, steel, white wood, PVC, and wood) with both wet and dry patches. Each frame is accompanied by a binary mask that highlights four roughly centralized saliva patches. We treat the first eight images as the tuning set and reserve the final two images for blind testing; the masks only cover the salient regions, so every method must operate in an unsupervised fashion while relying heavily on color, contrast, and spatial locality.

- **Imaging:** High-contrast UV-C photography with fluorescent saliva to accentuate deposits across heterogeneous materials.
- **Ground truth:** One binary mask per image that marks four foreground regions, which guide plausibility filters and evaluation.
- **Split:** Images 1-8 serve for parameter tuning, and 9-10 are used to report unseen performance.
- **Challenge:** The small dataset prevents supervised training, so all methods rely on heuristics shaped by saliva brightness and location.

2.2. Method A: MATLAB thresholding pipeline

Method A is based on MATLAB and its computer vision libraries. The idea behind this method is to create a binary mask that identifies saliva spots with high recall, using a simple and fast technique that utilizes resources already present in MATLAB. This method will convert the images to grayscale, enhance their contrast, train a model to create binary masks, and apply what it has learned to the test set.

2.2.1. Pre-processing

Luminance conversion The raw images, heavily red due to the UV-C setting, were imported to MATLAB and converted to luminance (black and white). The recommended ITU-R BT.601 weights to each colour channel ($0.299 \times \text{Red}$, $0.587 \times \text{Green}$ and $0.114 \times \text{Blue}$) were applied, yielding a visually correct grayscale image (Figures 1a and 1b).

CLAHE After this conversion, the images were treated with Contrast Limited Adaptive Histogram Equalization (CLAHE) [24] to enhance their contrast and facilitate the

identification of a suitable luminance threshold (Figure 1c). This method was used to obtain a locally optimized histogram equalization, while avoiding noise amplification.

2.2.2. Ground Truth evaluation

After treating the input image, the code turns to the corresponding ground truth file. It performs an analysis of the number of white pixels the ground truth possesses and divides this quantity by the number of white areas to obtain a plausible area range to search for when segmenting the input image. This step helps filter out elements with similar luminances but that are either more than 50% too small (noise) or too large (other image objects).

2.2.3. Training

Otsu bias tuning Otsu’s thresholding technique [14] automatically classifies pixels in a luminance image as foreground (white) or background (black) depending on a threshold. Its mathematical formulation calculates the threshold that better minimizes the intra-class variance between the pixels in each class (Figure 1d). Although it is a great method for images that generate a bimodal histogram (as those present in the dataset), it can suffer from sources with uneven lighting and end up underestimating the foreground pixels. To compensate for this effect, Method A employs a set of optimizations to produce masks that more accurately recreate the ground truth. The first one involves training a bias factor to be combined with the original threshold. This bias (Equation 1) lowers or increases Otsu’s threshold, resulting in more or fewer white pixels in the binary mask. A series of sweeps of values, ever reducing the search range, was performed to find the optimal bias.

$$t_{\text{new}} = t_{\text{otsu}} \times (1 - \text{bias}) \quad (1)$$

After this phase, a plausibility filter is used to maintain on the binary mask only the regions that have a plausible area as the saliva patches, and are roughly centralized. Their distance from the geometric center of the four ground truth saliva patches is measured, and closer regions are rewarded. After this filtering, the bias that maximizes the Dice Similarity Coefficient (DICE) between all binary masks from the training set and their respective ground truths is chosen. The DICE coefficient was chosen to combine precision and recall into a single, simple, and comparable measurement across methods.

2.2.4. Testing

With a defined threshold bias (Figure 2), images 9 and 10 were processed without reference to the ground truth, only being compared to it at the end to generate the precision/recall results. Otsu’s thresholding was applied, and the same two post-processing filters were adopted. The first step filtered out areas with a small pixel count to remove

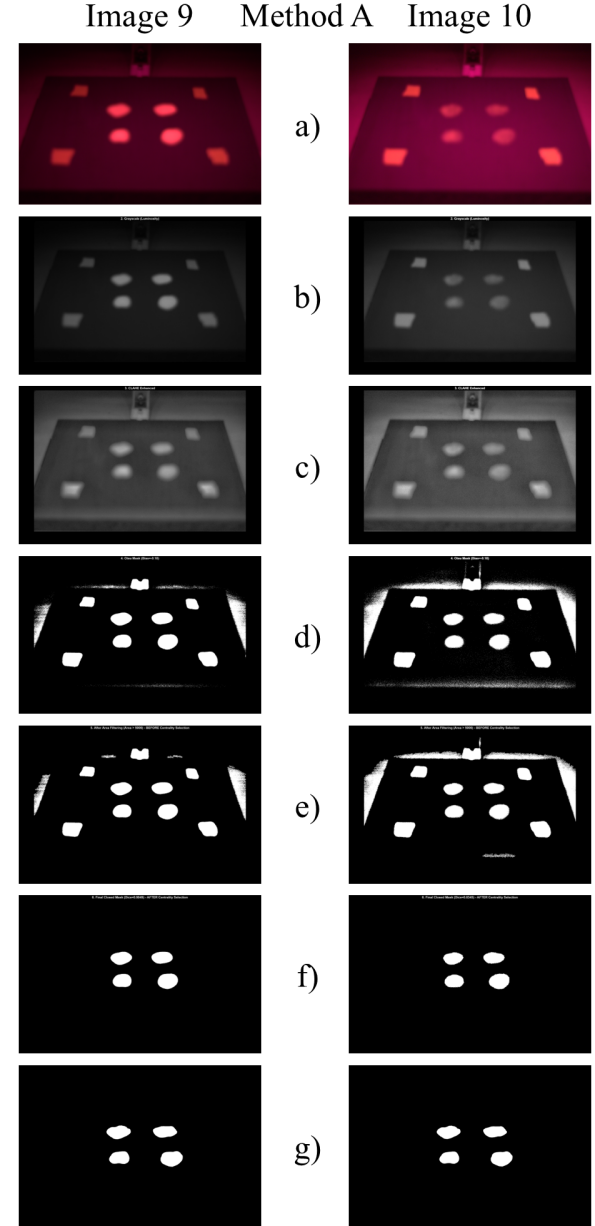


Figure 1. Step by step image processes done by Method A on the test images.

random noise and large areas that would not yield a feasible result for the binary mask (Figure 1e). The second filter is similar to the centrality logic applied during training, but since the ground truth images could not be consulted, centrality was measured from the center of the image. To finalize the mask, a Close operation (Erosion + Dilation) was performed to eliminate small gaps in the generated white areas (Figure 1f). The generated binary mask was then compared to the ground truth (Figure 1g) for performance eval-

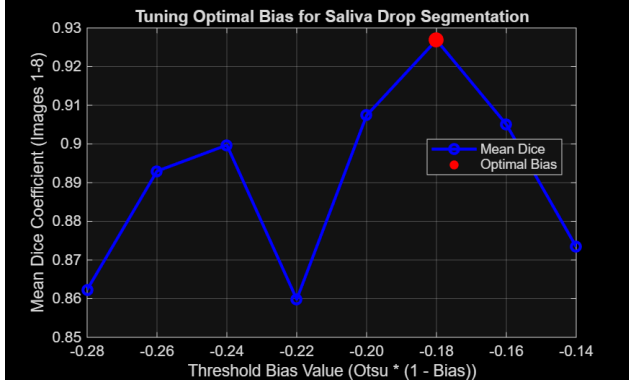


Figure 2. Bias parameter selected as -0.18 , meaning that the selected threshold was more permissive and normal.

uation.

2.3. Method B: Supervised per-channel thresholding and Linear Discriminant Analysis (LDA)

Method B establishes a supervised baseline using the 8 tuning images and their partial ground-truth masks. The goal is to estimate pixel-level decision rules that exploit the strong fluorescence contrast of saliva under UV-C illumination. We evaluate two approaches: (i) independent per-channel thresholds and (ii) a combined linear discriminant classifier.

2.3.1. Per-channel threshold learning

For each channel $c \in \{R, G, B\}$, we compute the distribution of pixel intensities for saliva and background using the labeled tuning subset. We sweep all unique intensity values as candidate thresholds and compute the Dice coefficient [6] obtained when predicting saliva whenever $x_c \geq t$. The optimal threshold for each channel is chosen as:

$$t_c = \arg \max_t \text{Dice}(t) \quad (2)$$

A pixel is classified as saliva if any channel exceeds its optimal threshold:

$$y_{R,G,B} = 1 \text{ if } R \geq t_R \vee G \geq t_G \vee B \geq t_B \quad (3)$$

This provides a simple supervised baseline in which each channel contributes independently.

2.3.2. Linear Discriminant Analysis (LDA)

While independent thresholds produce axis-aligned decision boundaries, the RGB channels are strongly correlated. As shown in Figure 3, saliva pixels exhibit extremely strong positive correlations across all three channels (0.97–0.99), meaning that their colour variation is almost entirely explained by a single underlying intensity dimension. In contrast, background pixels show weak or even negative cross-

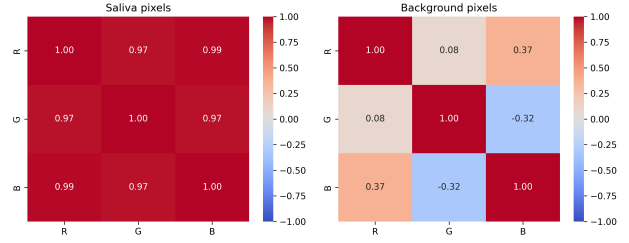


Figure 3. Correlation heatmap of RGB intensities for saliva and background pixels.

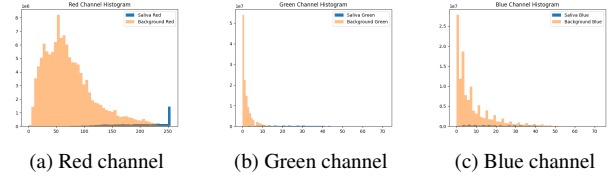


Figure 4. RGB channel histograms showing intensity distributions for saliva and background pixels.

channel correlations, indicating substantially greater chromatic variability.

The histograms illustrated in Figure 4 show the intensity distributions for each channel, demonstrating clear separation between saliva and background regions.

This difference motivates a multivariate classifier that can exploit joint RGB patterns rather than thresholding each channel independently.

We therefore learn a single discriminant direction $\mathbf{w} \in \mathbb{R}^3$ using Linear Discriminant Analysis (LDA) [8]. LDA computes a projection vector

$$\mathbf{w} = \mathbf{S}_w^{-1}(\boldsymbol{\mu}_{\text{saliva}} - \boldsymbol{\mu}_{\text{bg}}) \quad (4)$$

that maximizes the separation between saliva and background, where $\boldsymbol{\mu}_{\text{saliva}}$ and $\boldsymbol{\mu}_{\text{bg}}$ are the class means, and $\mathbf{S}_w$ is the pooled within-class covariance matrix.

Each pixel is mapped to a one-dimensional score $s = \mathbf{w}^T \mathbf{x}$.

We then sweep all unique score values on the tuning set and select the threshold T that maximizes the Dice coefficient [6].

The resulting classifier is

$$y = \begin{cases} 1 & \text{if } \mathbf{w}^T \mathbf{x} \geq T \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

This creates an oblique decision boundary in RGB space, enabling LDA to exploit correlated variations across channels and improving robustness relative to independent per-channel thresholds.

2.3.3. Pre- and Post-Processing

Before classification, a Gaussian blur is applied to suppress sensor noise while preserving the low-frequency structure of saliva fluorescence.

After thresholding, morphological closing and opening remove isolated noise pixels and fill small gaps, producing more coherent saliva regions.

2.4. Method C: exploratory k-means clustering

Method C looks at using K-Means clustering [10] to distinguish between saliva pixels and background pixels. K-Means clustering is an unsupervised machine learning algorithm used to partition a dataset into k distinct, non-overlapping subgroups (clusters). In the context of this image segmentation project, each pixel in the image is treated as a data point in a 3-dimensional color space (Red-Green-Blue).

The algorithm works iteratively to assign each data point to one of k groups based on feature similarity. The goal is to group pixels of similar color and intensity together. This is effective for saliva detection as the target (saliva) possesses distinct colorimetric properties (high brightness) compared to the background in each of the provided raw images.

2.4.1. Pre-processing

Standard image segmentation pipelines often begin with Gaussian blurring or Median filtering to reduce noise. However, for this specific dataset, no pre-processing filters were applied, as it was discovered that there was an overall improvement in Accuracy, Precision, and DICE Score results without filters (though there was a very slight decrease in Recall).

The tables showcasing these results can be found below. All results were created with $k = 3$ clusters.

Below table: *With pre-processing*

2.4.2. K-means

The K-means algorithm operates directly on RGB pixel values. To ensure robustness, the algorithm is executed 10 times with different random initializations, and the result with the best compactness metric (the smallest cluster variance) is selected. This multi-run approach mitigates the sensitivity of K-means to initial centroid placement, which can otherwise lead to suboptimal local minima.

After clustering, the center (centroid) of each cluster is computed. To identify which cluster corresponds to saliva, we calculate the perceived luminance for each cluster center using the standard RGB-to-luminance conversion: $Y = 0.114 \cdot B + 0.587 \cdot G + 0.299 \cdot R$. The cluster with the brightest center (highest luminance) is designated as the saliva cluster. Finally, all pixels belonging to the saliva cluster in the original image are assigned a value of 255, while all other pixels are set to 0, creating the binary segmentation mask.

Table 1. Pre-processing decision matrix for Method C.

Method	Action	Effect on Saliva Features	Result on Segmentation
Gaussian Blur	Smooths image by averaging pixels with neighbors.	Negative: Smears sharp specular highlights (the defining feature of wet saliva) into the darker background.	Under-segmentation: K-means centers converge on “gray” rather than “white,” missing the saliva boundaries.
Median Filter	Replaces pixel with median of neighbors.	Negative: Removes small “salt” noise. Since saliva glints often look like salt noise, this erases the target.	Data Loss: Small droplets are erased entirely.
Raw Input (Selected)	Direct input of pixel arrays.	Positive: Preserves peak luminance values (255, 255, 255) and sharp edges.	Optimal: Allows K-means to anchor a cluster specifically to high-intensity reflections.

Mathematical Foundation The algorithm aims to minimize the objective function J :

$$J = \sum_{j=1}^k \sum_{i=1}^n \|\mathbf{x}_i^{(j)} - \mathbf{c}_j\|^2 \quad (6)$$

Where:

- k is the number of clusters.
- $\mathbf{x}_i^{(j)}$ is a pixel point belonging to cluster j .
- $\mathbf{c}_j$ is the centroid (geometric center) of cluster j .

Algorithm Configuration The implementation uses `cv2.kmeans` with the following specific tuning to ensure robustness:

- **Criteria:** The algorithm stops if it reaches 100 iterations or if the centers move by less than $\epsilon = 0.2$ (convergence).
- **Stability (The ‘attempts’ parameter):** The code executes the K-means algorithm **10 distinct times** using different random initializations.
 - K-means is sensitive to the initial placement of centroids; a bad starting guess can lead to a “local minimum” (a suboptimal result).
 - By running it 10 times (`attempts=10`) and selecting the result with the best compactness, we ensure the segmentation is mathematically stable and repeatable.

Cluster Selection via Luminance Once the pixels are grouped into k clusters, we must determine which cluster represents “saliva.” We utilize a Luminance calculation based on the perception of brightness.

The code calculates the luminance (Y) for each cluster center using the standard coefficients for RGB (Red-Green-Blue) images:

$$Y = 0.114 \cdot B + 0.587 \cdot G + 0.299 \cdot R \quad (7)$$

We sort the calculated luminance values and select the cluster with the maximum luminance index as the positive class (Saliva). The logic behind this decision is that saliva is bright in each of the UV images, resulting in high pixel intensity.

2.4.3. Post-Processing (Morphological Filtering)

Raw K-means output often contains noise (isolated pixels) or gaps. We apply Morphological Transformations using a 5×5 kernel to refine the binary mask.

Morphological Opening:

- **Operation:** Erosion followed by Dilation.
- **Purpose:** This removes small, isolated specs of noise (false positives) from the background while keeping the larger saliva regions intact.

Morphological Closing:

- **Operation:** Dilation followed by Erosion.
- **Purpose:** This closes small black holes inside the white saliva regions (false negatives), ensuring the predicted saliva mass is solid and continuous.

2.4.4. Discussion

Experiments were conducted with $K = 2, 3, 4, 5$ clusters. The $K = 2$ configuration performed worst in terms of precision. The $K = 3$ configuration provides a good balance between precision and recall. While $K = 4$ also offers a reasonable balance, it begins to lose recall compared to $K = 3$. The $K = 5$ configuration loses too much recall to be practical.

The performance is evaluated using four metrics. Given the medical nature of the data (imbalanced classes where the background is much larger than the saliva), **Accuracy** is often misleading.

- **Accuracy:** Overall correctness. Usually high ($> 90\%$) because the vast majority of the image is background (True Negatives).
- **Precision:** High precision indicates we are not over-segmenting (classifying background as saliva).
- **Recall:** High recall means we aren’t missing subtle saliva traces.
- **Dice Score (F1):** The harmonic mean of Precision and Recall.

$$\text{Dice} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \quad (8)$$

Table 2. Method C ($K = 3$) metrics with pre-processing.

Image	Accuracy	Precision	Recall	Dice
1.tif	0.996	0.911	0.899	0.905
2.tif	0.838	0.089	0.821	0.161
3.tif	0.990	0.796	0.950	0.866
4.tif	0.959	0.428	0.896	0.579
5.tif	0.813	0.094	0.994	0.172
6.tif	0.778	0.104	0.984	0.188
7.tif	0.987	0.670	0.920	0.775
8.tif	0.947	0.377	0.979	0.544
9.tif	0.980	0.549	0.989	0.706
10.tif	0.962	0.352	0.833	0.495
AVERAGE	0.925	0.437	0.926	0.539

Table 3. Method C ($K = 3$) metrics without pre-processing (better result).

Image	Accuracy	Precision	Recall	Dice
1.tif	0.997	0.937	0.896	0.916
2.tif	0.844	0.092	0.820	0.165
3.tif	0.990	0.813	0.949	0.876
4.tif	0.960	0.432	0.895	0.583
5.tif	0.818	0.096	0.995	0.175
6.tif	0.779	0.105	0.985	0.190
7.tif	0.987	0.671	0.923	0.777
8.tif	0.949	0.387	0.977	0.554
9.tif	0.981	0.552	0.989	0.709
10.tif	0.963	0.358	0.820	0.498
AVERAGE	0.927	0.444	0.925	0.544

The Dice score provides the most balanced view of performance for segmentation tasks. A Dice score closer to 1.0 indicates a perfect overlap between the algorithm’s prediction and the ground truth.

Below table: *With pre-processing*

Below table: *Without pre-processing* (better result)

The rationale is that saliva in clinical images is characterized by “specular highlights”—sharp, high-frequency reflections of light. Blurring the image smears these high-intensity pixels into the darker surrounding tissue, reducing the contrast necessary for K-Means to distinguish the saliva cluster.

The following table outlines the comparative analysis that led to the exclusion of pre-processing filters:

2.5. Method D: probabilistic Gaussian mixture model

Method D upgrades Method C’s k-means clustering with a full Gaussian mixture model (GMM) that quantifies uncer-

tainty and chooses the most probable segmentation. Each pixel $\mathbf{x}_i = (R_i, G_i, B_i)^\top$ is modeled as a sample from a mixture of K Gaussian distributions:

$$p(\mathbf{x}_i | \Theta) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_i | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k),$$

where $K \in \{3, 4\}$, π_k are mixture weights ($\pi_k \geq 0$, $\sum_k \pi_k = 1$), $\boldsymbol{\mu}_k \in \mathbb{R}^3$ are mean RGB vectors, $\boldsymbol{\Sigma}_k \in \mathbb{R}^{3 \times 3}$ are full covariance matrices, and $\Theta = \{\pi_k, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}_{k=1}^K$ are all GMM parameters.

2.5.1. Pre-processing

The images remain in RGB space and are reshaped into pixel arrays so the GMM learns the native color distribution directly. Candidate means $\boldsymbol{\mu}_k^{(0)}$ are initialized using Method C's k-means centers, helping the Expectation-Maximization (EM) algorithm start near plausible saliva/background separations. No additional smoothing or filtering is applied, keeping preprocessing comparable to the other pipelines.

2.5.2. Expectation{Maximization algorithm

The parameters Θ are estimated via the Expectation-Maximization (EM) algorithm [4], implemented by `sklearn.mixture.GaussianMixture`:

1. E-step (responsibilities):

$$\gamma_{ik} = p(z_i = k | \mathbf{x}_i, \Theta^{(t)}) = \frac{\pi_k^{(t)} \mathcal{N}(\mathbf{x}_i | \boldsymbol{\mu}_k^{(t)}, \boldsymbol{\Sigma}_k^{(t)})}{\sum_{j=1}^K \pi_j^{(t)} \mathcal{N}(\mathbf{x}_i | \boldsymbol{\mu}_j^{(t)}, \boldsymbol{\Sigma}_j^{(t)})}$$

2. M-step (parameter update):

$$\pi_k^{(t+1)} = \frac{1}{N} \sum_{i=1}^N \gamma_{ik}, \quad (9)$$

$$\boldsymbol{\mu}_k^{(t+1)} = \frac{\sum_{i=1}^N \gamma_{ik} \mathbf{x}_i}{\sum_{i=1}^N \gamma_{ik}}, \quad (10)$$

$$\boldsymbol{\Sigma}_k^{(t+1)} = \frac{\sum_{i=1}^N \gamma_{ik} (\mathbf{x}_i - \boldsymbol{\mu}_k^{(t+1)}) (\mathbf{x}_i - \boldsymbol{\mu}_k^{(t+1)})^\top}{\sum_{i=1}^N \gamma_{ik}} \quad (11)$$

2.5.3. Multiple initializations and model selection

EM only guarantees convergence to a local optimum of the log-likelihood. To address this, for each image and each K , we run $n_{\text{runs}} = 10$ independent GMM fits, mixing k-means seeds with random starts to explore multiple basins of attraction.

For each run r , we compute the pseudo-log-likelihood (PLL):

$$\text{PLL}(\Theta_r) = \sum_{i=1}^N \log p(\mathbf{x}_i | \Theta_r) = N \cdot \frac{1}{N} \sum_{i=1}^N \log p(\mathbf{x}_i | \Theta_r),$$

which evaluates the model fit on the same data used for training. We select the best run:

$$r^* = \arg \max_r \text{PLL}(\Theta_r),$$

using Θ_{r^*} for the rest of the pipeline. This directly implements the principle of choosing the most probable solution.

2.5.4. From probability maps to binary masks

For the selected model Θ^* , EM yields responsibilities $\gamma_{ik} = p(z_i = k | \mathbf{x}_i, \Theta^*)$. To identify the saliva (foreground) component, we compute the luminance for each component mean $\boldsymbol{\mu}_k = (R_k, G_k, B_k)$:

$$Y_k = 0.299R_k + 0.587G_k + 0.114B_k,$$

and select $k_{\text{fg}} = \arg \max_k Y_k$ as the brightest component. The foreground probability map is then:

$$P_{\text{FG}}(\mathbf{x}_i) = \gamma_{ik_{\text{fg}}} = p(z_i = k_{\text{fg}} | \mathbf{x}_i, \Theta^*).$$

We threshold this probability map at $T = 0.5$ to create a binary mask:

$$M(i) = \begin{cases} 255, & P_{\text{FG}}(\mathbf{x}_i) > 0.5, \\ 0, & \text{otherwise.} \end{cases}$$

Finally, we apply morphological opening and closing using a 5×5 kernel to remove small noisy blobs and fill small holes, yielding the final cleaned saliva mask.

2.5.5. Results

Table 4 reports the average metrics across all images for $K = 3$ and $K = 4$. The probabilistic GMM achieves high recall (0.897 at $K = 3$, 0.630 at $K = 4$) by capturing bright saliva pixels that hard clustering sometimes misses. At $K = 3$, the model prioritizes recall over precision (0.272), resulting in many false positives due to Gaussian overlap. Increasing to $K = 4$ improves precision (0.320) and Dice (0.481) while maintaining reasonable recall, demonstrating how additional components provide tighter distributions and better separation between foreground and background. The PLL values (-2.92×10^6 at $K = 3$, -2.77×10^6 at $K = 4$) confirm that the $K = 4$ solution provides a better probabilistic fit to the RGB distributions.

Figure 5 illustrates representative GMM outputs for Image 1, comparing $K = 3$ and $K = 4$ configurations. The 3D RGB scatter plots show how the GMM components partition the color space, with the foreground (saliva) component marked by an 'X' and background components marked by circles. The binary masks demonstrate the final segmentation after thresholding the probability maps at 0.5 and applying morphological cleanup. The quantitative comparison shows that $K = 4$ achieves better precision and Dice

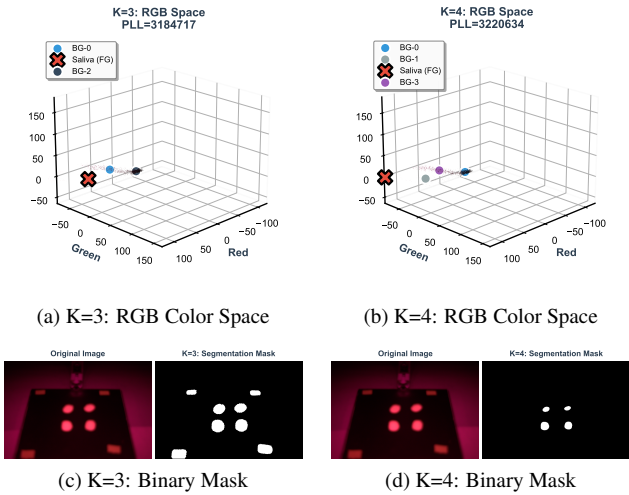


Figure 5. Method D (GMM) segmentation results for Image 1. Top row: 3D RGB scatter plots showing pixel distributions and GMM component centers for (a) $K=3$ and (b) $K=4$ configurations. The foreground (saliva) component is marked with an 'X', while background components are marked with circles. Bottom row: Binary masks showing the final segmentation for (c) $K=3$ and (d) $K=4$ after thresholding probability maps at 0.5 and morphological cleanup.

Table 4. Method D (GMM) average metrics for $K = 3$ and $K = 4$.

K	Accuracy	Precision	Recall	Dice	PLL (10^6)
3	0.939	0.272	0.897	0.426	-2.92
4	0.951	0.320	0.630	0.481	-2.77

scores while maintaining high recall, validating the model selection strategy based on pseudo-log-likelihood. This fully probabilistic workflow maintains the interpretability of k-means while adding soft assignments, likelihood-based selection, and quantified uncertainty via the posterior probability maps, making it particularly suitable for safety-critical applications where missing contamination is worse than false alarms.

3. Results and Discussion

Table 5 summarizes the average metrics across all evaluated configurations. Method C achieves the highest precision (0.569 at $K = 4$), while Method D’s GMM achieves high recall (0.897 at $K = 3$, 0.630 at $K = 4$). The $K = 4$ GMM configuration improves Dice (0.426 to 0.481) and PLL (-2.92 to -2.77×10^6), indicating a better probabilistic fit to the RGB distributions.

Method B establishes a supervised baseline using LDA, exploiting strong fluorescence contrast under UV-C illumi-

Method	K	Accuracy	Precision	Recall	Dice	PLL (10^6)
Method B (LDA, supervised)	—	—	—	—	—	—
Method C (k-means, no preprocessing)	3	0.927	0.444	0.925	0.544	—
Method C (k-means, no preprocessing)	4	0.953	0.569	0.817	0.601	—
Method D (GMM)	3	0.939	0.272	0.897	0.426	-2.92
Method D (GMM)	4	0.951	0.320	0.630	0.481	-2.77

Table 5. Cross-method comparison of segmentation metrics.

nation. Method C provides a fast hard-clustering baseline, with $K = 3$ offering balanced precision and recall, while $K = 4$ achieves higher precision (0.569) but lower recall (0.817). Method D’s probabilistic approach converts k-means centers into soft assignments, with PLL-based model selection identifying the best of ten EM runs. The probability maps recover gaps that hard clustering misses, achieving recall of 0.897 at $K = 3$ and improving Dice from 0.426 to 0.481 at $K = 4$.

Future work will expand the dataset and explore hybrid models, including supervised backbones such as U-Net or transformer encoders, while additional substrates and illumination settings will test generalization.

4. Conclusion

This report presents four segmentation methods for saliva detection under UV-C illumination: Method A’s MATLAB thresholding, Method B’s supervised LDA, Method C’s k-means clustering, and Method D’s likelihood-aware GMM. Method D achieves high recall while improving Dice at $K = 4$ through PLL-guided model selection, demonstrating that probabilistic modeling with multi-start EM delivers stable masks for safety-critical contamination imagery.

References

- [1] B. Allegranzi and D. Pittet. Role of hand hygiene in healthcare-associated infection prevention. *Journal of Hospital Infection*, 77(4):305–315, 2011. 1
- [2] B. Chang, R. J. Boltryk, Y. Amarouchene, and D. K. Dysthe. Aerial mucosalivary droplet dispersal distributions with implications for disease mitigation. *Physical Review Research*, 2(4):043391, 2020. 1, 2
- [3] N. F. de Oliveira Neto et al. The emergence of saliva as a diagnostic and prognostic tool for viral infections. *Viruses*, 16(11):1759, 2024. 1
- [4] A. P. Dempster, N. M. Laird, and D. B. Rubin. Maximum likelihood from incomplete data via the em algorithm. *Journal of the Royal Statistical Society: Series B (Methodological)*, 39(1):1–22, 1977. 7
- [5] C.-X. Deng, W. Ma, and Y. Yin. An edge detection approach of image fusion based on improved sobel operator. In *2011 4th International Congress on Image and Signal Processing*, pages 1934–1938, 2011. 2
- [6] L. R. Dice. Measures of the amount of ecologic association between species. *Ecology*, 26(3):297–302, 1945. 4

- [7] A. Dwivedi, S. Joshi, R. Kumar, V. Rai, V. Chaudhary, and P. K. Mishra. Optimizing diagnosis approaches: A computational perspective for medical image processing in cholangiocarcinoma detection. *Archives of Computational Methods in Engineering*, 2025. 2
- [8] R. A. Fisher. The use of multiple measurements in taxonomic problems. *Annals of Eugenics*, 7(2):179–188, 1936. 4
- [9] H. P. Loveday, J. A. Wilson, R. J. Pratt, et al. epic3: National evidence-based guidelines for preventing healthcare-associated infections in nhs hospitals in england. *Journal of Hospital Infection*, 86(Supplement 1):S1–S70, 2014. 1
- [10] J. MacQueen. Some methods for classification and analysis of multivariate observations. 1:281–297, 1967. 5
- [11] S. S. Magill, J. R. Edwards, W. Bamberg, et al. Multi-state point-prevalence survey of health care-associated infections. *New England Journal of Medicine*, 370(13):1198–1208, 2014. 1
- [12] D. Malamud. Saliva as a diagnostic fluid. *Dental Clinics of North America*, 55(1):159–178, 2011. 1
- [13] G. E. Miranda, J. M. Campos, and R. Pérez. Analysis of the fluorescence of body fluids on different surfaces for forensic applications. *Science & Justice*, 54:427–431, 2014. 1, 2
- [14] N. Otsu. A threshold selection method from gray-level histograms. *IEEE Transactions on Systems, Man, and Cybernetics*, 9(1):62–66, 1979. 3
- [15] J. A. Otter, S. Yezli, J. A. G. Salkeld, and G. L. French. Evidence that contaminated surfaces contribute to the transmission of hospital pathogens and an overview of strategies to address contaminated surfaces in hospital settings. *American Journal of Infection Control*, 44(5):S6–S11, 2013. 1
- [16] T.-H. Pham and K.-D. Nguyen. Enhanced droplet analysis using generative adversarial networks. *Computer Vision and Pattern Recognition, Cornell University*, 2024. 2
- [17] T. Rahman et al. Exploring the effect of image enhancement techniques on covid-19 detection using chest x-ray images. *Computers in Biology and Medicine*, 132:104319, 2021. 2
- [18] X. Ren and S. Lai. Medical image enhancement based on laplace transform, sobel operator and histogram equalization. *Academic Journal of Computing & Information Science*, 5(6), 2022. 2
- [19] A. Sarritzu et al. droplet-ml, 2024. <https://github.com/alesarritz/droplet-ml>. 2
- [20] S. H. Shah et al. Assessment of respiratory droplet transmission during the ophthalmic slit-lamp exam: A particle tracking analysis. *American Journal of Ophthalmology*, 222:76–81, 2021. 1, 2
- [21] K. K.-W. To, O. T.-Y. Tsang, C. C.-Y. Yip, et al. Consistent detection of 2019 novel coronavirus in saliva. *Clinical Infectious Diseases*, 71(15):841–843, 2020. 1
- [22] D. J. Weber and W. A. Rutala. Understanding and preventing transmission of healthcare-associated pathogens due to the contaminated hospital environment. *Infection Control & Hospital Epidemiology*, 34(5):449–452, 2013. 1
- [23] F. Wu, B. Zhang, J. Wu, R. Jia, and D. Sun. Quantitative measurement and application of droplets on physical surfaces based on laser-induced fluorescence. *Transport Safety and Environment*, 6(2):tdad030, 2024. 1, 2
- [24] K. Zuiderveld. Contrast limited adaptive histogram equalization. *Graphics Gems IV*, pages 474–485, 1994. 2

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